

# Epicure: Navigating the Emergent Geometry of Food Ingredient Embeddings

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## Abstract

FlavorGraph [Park et al., 2021] is the most comprehensive public food embedding to date, combining FlavorDB chemistry with Recipe1M+ co-occurrence into a single Metapath2Vec model. In earlier work we showed that FlavorGraph’s 300-D embeddings already encode at least fifteen interpretable culinary dimensions – spanning taste, texture, nutrition, geography, culture, and processing – and that LLM-augmented vocabulary consolidation strengthens most of those signals [Radzikowski and Chen, 2026]. That study was tied to a single English-centric pretraining, however, and fused chemical and recipe-context signal as a fixed inductive bias rather than a controllable design axis. We present **Epicure**, a family of three sibling skip-gram ingredient embeddings retrained from scratch on a multilingual recipe corpus.

We aggregate 4.14M recipes from 11 sources spanning seven languages (English, Chinese, Russian, Vietnamese, Spanish, Turkish, Indonesian, German, and Indian-English) and normalise the raw ingredient strings to 1,790 canonical entries via an LLM-augmented pipeline. A 203,508-edge ingredient–ingredient NPMI graph and an 80,019-edge typed FlavorDB ingredient–compound graph (2,247 typed compound nodes across 15 categories) seed three Metapath2Vec variants that share architecture and hyperparameters and differ only in the random-walk schema: **Cooc** walks the co-occurrence graph only, **Chem** walks the typed compound metapaths only, and **Core** blends both via injected ingredient–ingredient walks at controlled mixing, placing each model at a distinct point on the chemistry-vs-recipe-context spectrum.

All three Epicure models linearly recover supervised probes: 27 continuous sensory and nutrient directions and 8 cuisine macro-regions, with mean Cohen’s  $d$  for cuisine separability of 2.43/2.70/3.07 for Cooc/Core/Chem. An unsupervised multi-seed-stable FastICA decomposition on food-group-residualised embeddings recovers 20 interpretable factors per model, and a Gaussian-mixture-model (GMM) partition of each factor’s high-quartile yields 150–200 named culinary modes per model with mean coherence 0.611/0.833/0.703 against random-pair baselines of 0.097/0.348/0.115. Two complementary operator families run on the same 300-D embedding: nearest-neighbour *pairings* (top- $K$  and mode-membership lookups) and SLERP *direction arithmetic* that rotates a seed toward either a supervised pole vector (*rice* + South-Asian  $\rightarrow$  *curry leaf*, *urad dal*, *chana dal*, *fenugreek seed*) or an emergent factor-mode pole, controlled by a continuous angle  $\theta$  that interpolates between seed-dominated and target-dominated retrieval.

The three sibling embeddings make chemistry-vs-recipe-context a controllable design axis at the walk schema and expose both label-grounded and emergent navigation operators on a single 300-D space, supporting chef-facing tools that can rotate, blend, or retrieve along either supervised semantic directions or culturally-coherent emergent modes. Code and trained artefacts are not released at this time.

## 1 Introduction

A chef asked what pairs with *miso* reaches for *mirin*, *dashi*, or *sesame oil*. Asked what pairs with *olive oil*, they reach for *basil*, *tomato*, or *prosciutto*. Such choices are knowledge embedded

in recipe corpora across cultures and embodied in the working intuition of cooks and chefs. A computational representation of this knowledge would enable a class of downstream tools: menu and recipe assistants that surface plausible companions for an ingredient on hand; cross-cuisine navigation that lets a Mediterranean seed find its East-Asian peers without manual lookup; and sensory- or nutrient-aware exploration that places an ingredient inside an interpretable axis (fatty, fermented, bitter, high-protein). A useful ingredient embedding model is the substrate for all of these.

Computational gastronomy has approached this target from two complementary directions. [Ahn et al. \[2011\]](#) introduced the *flavour network* and established cultural divergence in compound-sharing as an empirical phenomenon. [Garg et al. \[2017\]](#) catalogued the aroma molecules of 936 food entities (FlavorDB), and [The Metabolomics Innovation Centre \[2020\]](#) extended chemical coverage to FoodDB’s 70,000 compounds. These chemical resources underpin FlavorGraph [\[Park et al., 2021\]](#), which combined FlavorDB with Recipe1M+ [\[Marin et al., 2021, Salvador et al., 2017\]](#) in a heterogeneous graph of 6,653 ingredients and 1,645 compounds trained with Metapath2Vec; it is the most comprehensive public food embedding to date. Symbolic alternatives such as FoodKG [\[Hausmann et al., 2019\]](#) integrate recipe, nutrition, and ontology data into RDF knowledge graphs targeted at recommendation.

A separate line of work studies what kinds of computations a dense embedding actually supports, and this paper draws directly on three of its findings. [Mikolov et al. \[2013\]](#) established that semantic relationships emerge as linear directions in word2vec ( $king - man + woman = queen$ ); the directional view underwrites both our 27 supervised culinary probes (cuisine, food-group, NOVA, USDA macronutrients, sensory) and the 20 unsupervised FastICA factors we recover per model, together with the SLERP rotation operator that traverses any of them continuously. [Mu et al. \[2017\]](#) argued that embedding isotropy is a precondition for stable directional operations and proposed post-hoc rescue methods (all-but-the-top, whitening) for collapsed geometries; we measure isotropy directly via participation ratio and average pairwise cosine and find that our three siblings sit at sharply different points on that spectrum – a property of the walk schema rather than the input data. [Caliskan et al. \[2017\]](#)’s Word Embedding Association Test (WEAT) provides the standard diagnostic for whether named semantic axes are reflected in the geometry; we report it in the supplement alongside other robustness checks.

In earlier work [\[Radzikowski and Chen, 2026\]](#) we analysed FlavorGraph’s 300-D embeddings and found at least fifteen interpretable culinary dimensions – spanning taste, texture, nutrition, geography, culture, and processing – with LLM-augmented vocabulary consolidation strengthening most of those signals. That analysis was bounded by FlavorGraph’s fixed pretraining on three counts: a single English-centric corpus, a single mix of chemistry and recipe-context signal, and a scattered ingredient vocabulary that included preparation details and non-food items.

We present **Epicure**, a family of three skip-gram ingredient embeddings retrained from scratch to lift those three bounds simultaneously. We aggregate a 4.14M-recipe multi-language corpus (English, Chinese, Russian, Vietnamese, Spanish, Turkish, Indonesian, German, and Indian-English), normalise it to a shared 1,790-ingredient LLM-curated canonical vocabulary, and expose the chemistry-vs-recipe-context mix as a controllable design axis at the walk schema. All three siblings share architecture and hyperparameters; they differ only in which random walks the skip-gram objective sees: **Cooc** walks recipe co-occurrence, **Chem** walks typed FlavorDB compound-ingredient metapaths, and **Core** blends both via injected ingredient-ingredient walks at controlled mixing. The three siblings thus trace the chemistry-vs-recipe-context spectrum from a single experimental design.

In the trained embeddings, supervised directions for cuisine, food-group, NOVA processing class, USDA macronutrients, and 19 sensory categories are linearly recoverable; an unsupervised multi-seed-stable FastICA decomposition on top of the embeddings rediscovers 20 interpretable axes per

model, and Gaussian-mixture-model (GMM) partitioning of each factor’s high-quartile yields 150–200 named culinary modes per model. The supervised and emergent geometries together expose two operator families on the same 300-D embedding: nearest-neighbour *pairings* (top- $K$  neighbours plus mode-membership lookup) and SLERP *direction arithmetic* that rotates a seed toward either a supervised pole vector or an emergent factor-mode pole.

## 2 Methods

The pipeline runs in five stages: (i) aggregate a multilingual recipe corpus, (ii) normalise the raw NER terms into a canonical ingredient vocabulary, (iii) construct the co-occurrence and typed-compound graphs, (iv) train three Metapath2Vec variants (Cooc, Core, Chem) on those graphs, and (v) analyse the resulting embeddings with supervised direction probes and unsupervised factor / mode discovery.

### 2.1 Corpus

We aggregate recipes from 11 publicly available datasets spanning seven languages, yielding 4,135,189 recipes dominated by the English RecipeNLG [Bień et al., 2020] (53.9%) and the Chinese XiaChuFang [Liu et al., 2022] (37.4%) corpora, with the Russian Povarenok [Rogozinushka, 2021] corpus contributing 3.5% and eight smaller multilingual corpora covering Vietnamese [Nguyen, 2024], Spanish [SomosNLP, 2023b, Frorozco, 2023, SomosNLP, 2023a], Turkish [Al, 2023], Indian (in English) [Jain, 2020, Singh, 2019, Ahsan, 2022], Indonesian [Dzikri, 2020], and German [Sterby, 2021]. Per-source recipe counts and macro-region backing are catalogued in the supplement’s *Corpus and Vocabulary* appendix. Non-English ingredient terms are machine-translated to English by the Claude Opus family (internal deployment ID 4.6) [Anthropic, 2026a] under deterministic decoding (temperature 0); after merging, deduplication, and intersecting with the final 1,790-canonical vocabulary, 4,103,118 recipes (99.2%) contain at least one matched ingredient, with recipes carrying fewer than two matches contributing no co-occurrence pair to the NPMI step.

### 2.2 Canonical Vocabulary

Raw named-entity-recognition (NER) extraction across all eleven sources yields roughly  $\sim 200,000$  unique ingredient strings, dominated by spelling variants, brand names, non-food items, and preparation modifiers. An LLM-augmented canonicalisation pipeline uses the Claude Opus family (internal deployment ID 4.6) [Anthropic, 2026a] with deterministic decoding for term classification and Gemini Embedding models for semantic clustering [Lee et al., 2025]. Production dedup runs used Google’s API model identifier `gemini-embedding-001` [Google Cloud, 2026], followed by a final manual curation pass. This reduces the set to **1,790 canonical ingredients**. Ingredient matching to FlavorDB [Garg et al., 2017] follows an entity-unique policy: each FlavorDB entity matches at most one canonical ingredient, with name-similarity tiebreaking when several candidates compete (the supplement’s *Graph Construction* appendix details the policy). After graph construction, 523 ingredients retain active typed I–C edges after the `min_compound_degree=2` filter applied during graph construction; the remaining 1,267 are non-hub. Nutrient and sensory labels are matched against USDA FoodData Central [U.S. Department of Agriculture, Agricultural Research Service, 2019] and FlavorDB. The canonical-vocabulary CSV pairs each normalised ingredient name with its FlavorDB and USDA anchors. Principal counted sets used throughout the paper are summarised in Table 1.

Table 1: Principal counted sets in the normalisation and evaluation pipeline.

| Counted set                                          | $n$   |
|------------------------------------------------------|-------|
| Final canonicals (embedded vocabulary)               | 1,790 |
| Cuisine-tagged total (pre co-occurrence filter)      | 1,816 |
| – Universal (no distinctive region)                  | 808   |
| – Cuisine-specific                                   | 1,008 |
| • single-label                                       | 858   |
| • multi-label                                        | 150   |
| Cuisine-clustering subset (specific $\cap$ embedded) | 986   |
| Food-group-clustering subset (USDA $\cap$ embedded)  | 1,560 |

**Cuisine taxonomy.** For cuisine evaluation we define eight macro-regional cuisine clusters grounded in corpus provenance (Table 2). Claude Opus family models (internal deployment ID 4.6) [Anthropic, 2026a] tag every canonical ingredient with zero or more macro-region labels under a *distinctive-marker* prompt: universal ingredients (salt, onion, egg, flour, rice) are left untagged, and only ingredients that immediately signal a culinary tradition receive a region label. Of 1,816 tagged ingredients, 808 are universal (44.5%) and 1,008 are cuisine-specific (55.5%); intersected with the final 1,790-canonical embedded set this yields 986 ingredients for cuisine-clustering evaluation, of which 858 carry a single region label and 150 carry two or three.

Table 2: Eight cuisine macro-regions and their approximate recipe-count backing in the training corpus.

| Macro-region     | Backing recipes | Constituent traditions                                             |
|------------------|-----------------|--------------------------------------------------------------------|
| East Asian       | 1,549,034       | Chinese, Korean                                                    |
| Western Atlantic | 198,086         | American, British, German, Scandinavian                            |
| Mediterranean    | 164,107         | Italian, French, Iberian, Greek, Levantine, North African, Turkish |
| Eastern European | 154,479         | Russian, Ukrainian, Polish, Hungarian, Georgian                    |
| Southeast Asian  | 107,964         | Thai, Vietnamese, Filipino, Indonesian, Malay                      |
| South Asian      | 47,462          | Indian, Pakistani, Sri Lankan, Bangladeshi                         |
| Latin American   | 40,618          | Mexican, Caribbean, Brazilian, Peruvian, Colombian                 |
| Japanese         | 33,923          | Japanese                                                           |

## 2.3 Graph Construction

The three Epicure models share the same 1,790-ingredient node set and the same 203,508 NPMI co-occurrence edges (Table 3). Two graph variants are constructed:

**Cooc graph (co-occurrence only).** Ingredient–ingredient edges weighted by normalised point-wise mutual information [Bouma, 2009] (NPMI) computed over the 4.10M matched recipes. Ingredients appearing in fewer than 20 recipes are dropped before NPMI computation, which together with the canonicalisation pipeline yields the 1,790-ingredient vocabulary. After retaining only positive-NPMI pairs, the graph has 203,508 edges.

**Core/Chem graph (co-occurrence + typed compound edges).** Adds 2,247 typed FlavorDB compound nodes connected to ingredient nodes by 80,019 typed I–C edges. Each original compound carries one or more of 15 flavor-category tags (balsamic, citrus, earthy, fatty, floral, fruity, green,

meaty, minty, nutty, spicy, vegetable, wine-like, woody, plus one residual); compounds are replicated once per category they belong to so Metapath2Vec’s typed walks can distinguish a citrus–citrus compound overlap from a citrus–earthy bridge. This approach differs from FlavorGraph’s single-type compound node, which is a single node for all compounds of a given type.

Table 3: Graph variant statistics. Cooc operates on a pure ingredient–ingredient graph; Core and Chem share an identical heterogeneous graph that adds 2,247 typed FlavorDB compound nodes and 80,019 typed I–C edges. All three models share the same 1,790-ingredient vocabulary and 203,508 NPMI co-occurrence edges; the difference between Core and Chem is the walk schema, not the graph.

|                            | Cooc graph | Core/Chem graph |         |
|----------------------------|------------|-----------------|---------|
|                            | (Cooc)     | (Core)          | (Chem)  |
| Ingredient nodes           | 1,790      | 1,790           | 1,790   |
| Compound nodes (typed)     | —          | 2,247           | 2,247   |
| I–I edges (NPMI > 0)       | 203,508    | 203,508         | 203,508 |
| I–C edges (typed)          | —          | 80,019          | 80,019  |
| Total typed graph edges    | 203,508    | 283,527         | 283,527 |
| Compound types             | —          | 15              | 15      |
| Ingredients with I–C edges | —          | 523             | 523     |

## 2.4 The Three Epicure Models

We train three metapath2vec [Dong et al., 2017] models with identical architecture and hyperparameters (Table 4: 300-dim embeddings, `walks_per_node=100`, `walk_length=50`, `context_size=7`, 5 negative samples, `batch_size=32,768`, `lr=0.0025`, 20 epochs, no warm restart).

The objective is skip-gram with negative sampling [Mikolov et al., 2013].

Implementation uses the PyTorch framework [Paszke et al., 2019]. We refer to the family collectively as **Epicure** and to its three siblings as Epicure-Cooc, Epicure-Core, and Epicure-Chem. They differ only in which random walks the skip-gram objective sees:

**Epicure-Cooc.** Walks the Cooc graph: pure I–I random walks weighted by NPMI. No compound nodes.

**Epicure-Core.** Walks the typed-compound graph and injects pure I–I walks at `-ii_repeat=10` alongside the typed-compound metapaths. Edge transitions are weighted so I–C hops are not oversampled relative to the smaller I–I edge set. The resulting embedding blends chemical and recipe-context signal.

**Epicure-Chem.** Walks the typed-compound graph but with `-ii_repeat=0`: the I–I templates are absent and the only walks the skip-gram sees are compound-mediated. The chemistry extreme of the family.

The three models trace a chemistry-vs-recipe-context walk-template spectrum from a single experimental design. Section 3 characterises how this spectrum manifests in the trained embeddings; Section 4 exploits it.

Table 4: Metapath2Vec training hyperparameters. All three Epicure models share every architecture and optimiser setting; the bottom block lists the walk-design choices that differ across the three variants. All runs use PyTorch with SparseAdam and a fixed seed.

| Parameter                                     | Cooc              | Core              | Chem       |
|-----------------------------------------------|-------------------|-------------------|------------|
| Embedding dimension                           | 300               | 300               | 300        |
| Walks per node                                | 100               | 100               | 100        |
| Walk length                                   | 50                | 50                | 50         |
| Context window                                | 7                 | 7                 | 7          |
| Negative samples                              | 5                 | 5                 | 5          |
| Batch size                                    | 32,768            | 32,768            | 32,768     |
| Learning rate                                 | 0.0025            | 0.0025            | 0.0025     |
| Optimiser                                     | SparseAdam        | SparseAdam        | SparseAdam |
| LR schedule                                   | constant          | constant          | constant   |
| Epochs                                        | 20                | 20                | 20         |
| Random seed                                   | 42                | 42                | 42         |
| <i>Walk design</i>                            |                   |                   |            |
| Graph backbone                                | Cooc              | Core/Chem (typed) |            |
| Typed compound walks                          | —                 | yes               | yes        |
| Weighted I-C transitions                      | —                 | yes               | yes        |
| I-I walk injection ( <code>ii_repeat</code> ) | native (only I-I) | 10×               | 0 (none)   |

**Walk metapaths in detail.** Compounds attach only to the 523 ingredients with active I-C edges (§2.2); following the FlavorGraph nomenclature [Park et al., 2021], we call these *chemical-hub* (H) ingredients and the remaining 1,267 *non-hub* (N) ingredients. With  $C[x]$  denoting a compound of family  $x$ , Core and Chem generate three families of typed-compound walks, each playing a distinct role: *within-type*  $H-C[x]-H$  aggregates ingredient pairs that share a same-family compound; *via-compound*  $N-H-C[x]-H-N$  is the only route by which non-hub ingredients receive compound-mediated context; and *cross-type*  $C[x]-H-N-H-C[y]$  bridges two compound families through a hub-non-hub ingredient chain. Each of the 15 compound types receives one within-type and one via-compound template;  $2n = 30$  cross-type templates are sampled per walk round with a coverage guarantee that every type appears as both source and target. Core additionally samples ten pure I-I templates per walk round, so the I-I context is roughly an order of magnitude more frequent than any single compound-mediated template. We cycle templates with naive `pos % len(template)`, which deviates from FlavorGraph’s palindromic convention and concentrates the chemistry signal into short, high-information walks; an ablation in the supplement’s *Walk schema cycling* subsection documents the resulting walk-length distribution and a side-by-side comparison against the palindromic alternative.

## 2.5 Evaluation

The trained embeddings are evaluated under three blocks that map 1:1 onto the geometry section that follows.

**Direction quality.** We score 27 continuous probes and 8 cuisine macro-regions, all intersected with the three models’ shared vocabulary, under 5-fold repeated cross-validation. Continuous probes report Spearman  $\rho$  between an ingredient’s projection onto a fold-trained linear direction and its ground-truth score; cuisine probes report one-vs-rest Cohen’s  $d$  on the distinctive-marker tags. The continuous probes are organised into three strata that progressively decouple from the typed I-C walk schema: 14 baked-in compound-feature (CF) sensory categories the schema sees directly (e.g.



cf\_citrus), 5 held-out basic-taste CF probes the schema does not see, and 8 USDA macronutrient probes drawn from external nutrient data (e.g. usda\_protein\_g); the 8 cuisine macro-regions, drawn from LLM-annotated distinctive-marker tags (e.g. Japanese), form a fourth stratum further removed from the training signal. Stratum design and the regression protocol are detailed in the supplement’s *Stratified Direction Quality* appendix.

**Intrinsic geometry.** Participation ratio (PR) and average pairwise cosine quantify isotropy. Normalised mutual information (NMI) measures self-organisation around 17 USDA food groups (single-label) and 8 cuisine macro-regions (multi-label); the soft-NMI variant used for the cuisine case is defined in the supplement’s *Multi-label NMI protocols* subsection. Silhouette and  $k$ NN@5 purity are reported as auxiliary cluster-quality metrics in Table 5.

**Emergent geometry.** 20 ICA factors are extracted per model with `sklearn.FastICA` [Pedregosa et al., 2011] on the *food-group-residualised* embedding so the recovered axes are orthogonal to the dominant food-group variance. Factor identifiability is enforced via Hungarian matching across 10 random seeds; the seed whose components have the highest mean matched-cosine across the others is retained, factors are sorted by stability descending (so factor index 0 is the most reproducible), and only factors with split-half cosine stability above 0.6 are kept (supplement’s *Multi-seed FastICA protocol* subsection). For each ICA factor, the top-quartile ingredients are partitioned in PCA-reduced space into Gaussian-mixture-model modes under BIC over  $K \in \{3, \dots, 7\}$  with a six-member minimum per mode; each resulting mode is projected back to 300-D as a unit-mean “pole”. The same GMM procedure is run in parallel on the high-quartile of every property in a curated supervised set (NOVA processing level, CF/USDA/LLM sensory scores, food-group binaries) so that emergent factor modes and supervised-property modes share the same representation. Mode coherence is the mean within-mode pairwise cosine, baselined against random-pair samples of the same size.

### 3 Geometry

We characterise the three Epicure embeddings in three steps: isotropy and food-group separation (Section 3.1) quantify how broadly each model spreads variance and how cleanly food groups separate; supervised direction quality (Section 3.2) measures how well linear directions recover labelled probes; emergent geometry (Section 3.3) reports the unsupervised ICA- $n=20$  factor analysis and the GMM modes that fall out of it, plus a coherence metric quantifying how tight the modes are. The 20 ICA factors and 150–200 modes per model are the geometric vocabulary that the operators in Section 4 act on.

#### 3.1 Isotropy and food-group structure

In order to characterise the basic conditioning of each embedding before testing linear operators on it, we measured two intrinsic geometry diagnostics (participation ratio, average pairwise cosine) and two unsupervised label-recovery diagnostics (normalised mutual information against the 17 USDA-derived food groups and against the eight cuisine macro-regions).

We found two isotropic geometries and one concentrated one: Cooc reaches participation ratio  $PR = 173.6$  of 300 possible dimensions and Chem  $PR = 183.1$ , both with average pairwise cosine in the 0.10–0.12 band, while Core sits at  $PR = 94.2$  with average pairwise cosine 0.35. This means the concentration in Core is a property of its walk schema rather than its inputs: Core injects

Table 5: Intrinsic geometry of the three Epicure models. Higher participation ratio (PR) and lower average pairwise cosine indicate a more isotropic embedding. Cuisine NMI is the soft multi-label variant. Clustering metrics use the common labelled subset ( $n=1,560$  food-group,  $n=986$  cuisine); bracketed values are 95% CIs from  $0.8n$  subsample bootstrap, 200 iterations. **Bold** marks the best result per row across the three models.

| Category                                                               | Metric                            | Cooc                        | Core                         | Chem                           |
|------------------------------------------------------------------------|-----------------------------------|-----------------------------|------------------------------|--------------------------------|
| <i>Isotropy</i>                                                        |                                   |                             |                              |                                |
|                                                                        | $N$ ingredients                   | 1,790                       | 1,790                        | 1,790                          |
|                                                                        | Participation ratio $\uparrow$    | 173.6                       | 94.2                         | <b>183.1</b>                   |
|                                                                        | Avg. pairwise cosine $\downarrow$ | <b>0.099</b>                | 0.349                        | 0.117                          |
|                                                                        | PCA top-10 variance               | <b>0.138</b>                | 0.234                        | 0.141                          |
|                                                                        | PCA top-50 variance               | <b>0.301</b>                | 0.434                        | 0.360                          |
| <i>Food-group clustering (17 categories, <math>n=1,560</math>)</i>     |                                   |                             |                              |                                |
|                                                                        | NMI                               | 0.205 [0.191, 0.223]        | <b>0.235</b> [0.212, 0.251]  | 0.226 [0.215, 0.248]           |
|                                                                        | $k$ NN@5 purity                   | 0.307 [0.282, 0.307]        | 0.352 [0.328, 0.349]         | <b>0.355</b> [0.332, 0.353]    |
|                                                                        | Silhouette                        | -0.036 [-0.045, -0.028]     | -0.053 [-0.062, -0.037]      | <b>-0.028</b> [-0.038, -0.015] |
| <i>Cuisine-region clustering (8 macro-regions, <math>n=986</math>)</i> |                                   |                             |                              |                                |
|                                                                        | Soft NMI                          | <b>0.457</b> [0.402, 0.481] | 0.456 [0.441, 0.509]         | 0.432 [0.399, 0.479]           |
|                                                                        | $k$ NN@5 Jaccard purity           | 0.652 [0.630, 0.656]        | <b>0.695</b> [0.670, 0.697]  | 0.677 [0.654, 0.680]           |
|                                                                        | Silhouette                        | +0.028 [0.026, 0.030]       | <b>+0.050</b> [0.045, 0.054] | +0.039 [0.036, 0.041]          |

each ingredient–ingredient edge as a length-2 walk and repeats those injected walks ten times per round (Section 2), creating strong recipe-context attractors; Cooc lacks the typed I–C metapaths and Chem lacks the injected I–I repetition, so both end up similarly spread.

We also found that all three embeddings organise themselves around nutritional and cultural labels without those labels being used for training: ingredients from the same USDA food group land closer together than chance, scoring 0.20–0.25 on normalised mutual information (NMI; 0 = chance, 1 = perfect recovery), and soft NMI on the eight cuisine macro-regions rises to 0.43–0.46 across the three models – roughly double the food-group level. This means cultural tradition shapes ingredient co-occurrence more cleanly than nutritional category. Figure 1 visualises the cuisine structure: a 2-D UMAP projection of each model coloured by cuisine macro-region surfaces visibly distinct East Asian, South Asian, Latin American, and Mediterranean clusters in all three Epicure variants.

Both label structures appear without any supervision; whether they translate into usable linear directions is the next question.





Figure 1: 2-D UMAP projection (cosine, `n_neighbors=30`, `min_dist=0.03`) of each Epicure model’s 1,790 ingredients, coloured by cuisine macro-region; universally tagged ingredients are de-emphasised in grey so the cultural structure dominates visually. All three models exhibit clearly separated East Asian, South Asian, Latin American, and Mediterranean clusters, with the tightness of those regions paralleling the isotropy ordering: Core’s compressed geometry compresses the clusters as well, while the isotropic Cooc and Chem produce more diffuse but still cleanly partitioned regions. The same UMAP coordinates, coloured by USDA food group, are reproduced in the supplement’s *UMAP Visualisations* appendix.

### 3.2 Direction quality

In order to test whether labelled culinary concepts are linearly recoverable in each embedding – and how that recoverability varies as the probe decouples from the training signal – we ran the five-fold cross-validated direction-quality protocol of Section 2.5 on the four-stratum probe set (14 baked-in CF + 5 held-out basic-taste CF + 8 USDA macros + 8 cuisine macro-regions).

We found that all three models recover every stratum linearly, with the same ordering Cooc < Core < Chem at each one: baked-in CF  $\bar{\rho} = 0.28/0.40/0.46$ ; held-out basic-taste CF 0.32/0.42/0.47; USDA macros 0.41/0.45/0.49; cuisine macro-regions  $\bar{d} = 2.43/2.70/3.07$ . Across the 27 continuous probes Chem beats Core on 26 and Cooc on 27, and leads on 8 of 8 cuisine regions. This means linear directions are usable navigation primitives in all three siblings, and the chemistry-heavy walk schema (Chem) sharpens them most – complementing rather than overriding the recipe-context signal. The supplement’s *Stratified Direction Quality* appendix reports stratum-level robustness checks, including an orthogonal-residual SNR ranking,  $\ell_1$ -regularised linear probes on categorical and continuous targets, and a held-out cross-modal validation against external FlavorDB and USDA labels.

Supervised directions answer “where labelled concepts live”; the embedding’s own natural axes need not coincide with any label, which the next subsection takes up.

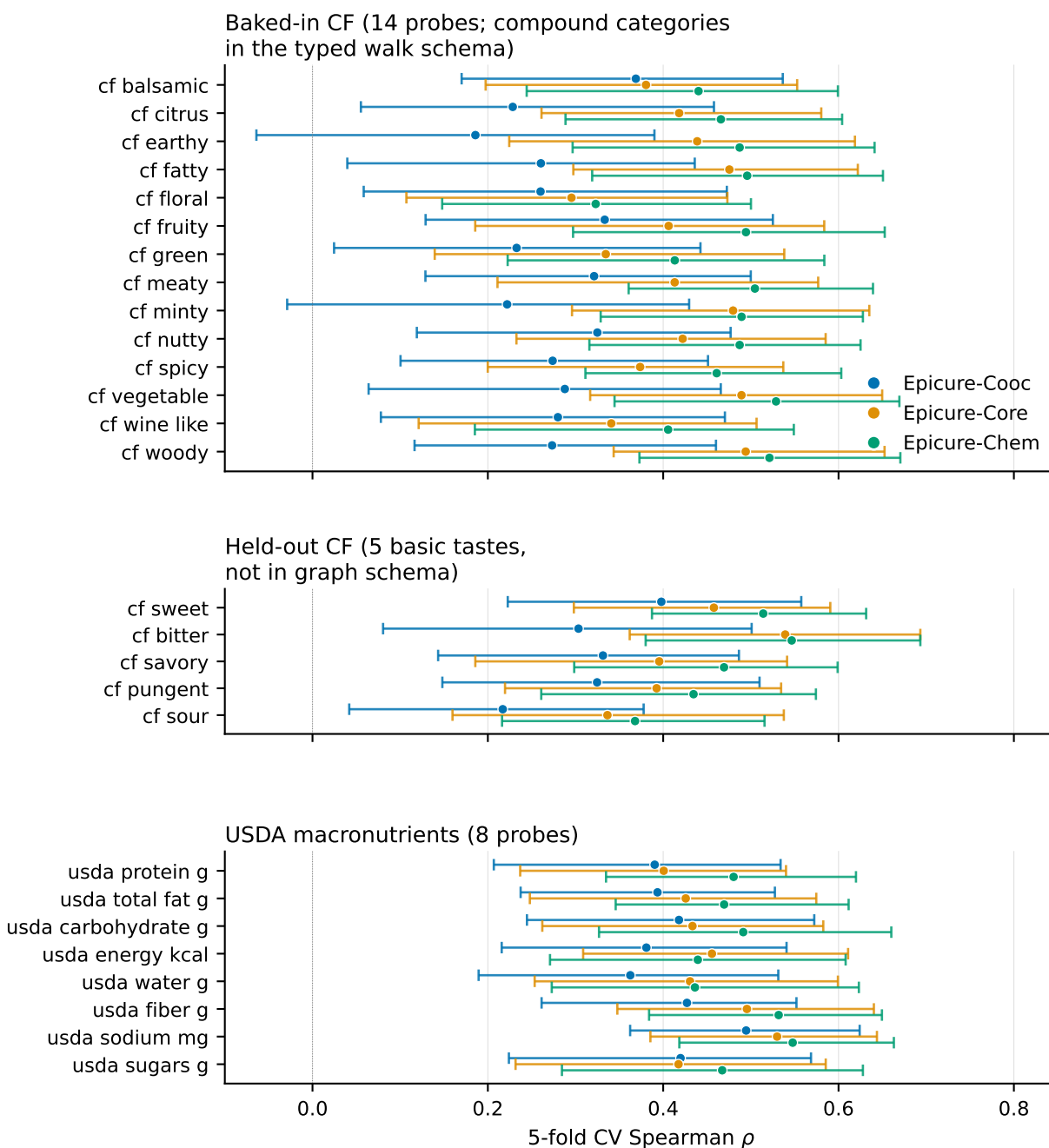


Figure 2: Direction quality as 5-fold repeated cross-validated Spearman  $\rho$  between each ingredient’s projection onto the linear direction (positive vs. negative pole separation) and its ground-truth score, with point estimate and 95% CI per Epicure model. The 27 continuous probes split into three strata: 14 FlavorDB compound-feature (CF) sensory categories whose labels index Core’s and Chem’s typed I–C walk schema (e.g. `cf_citrus`); 5 basic-taste CF probes outside the graph schema; and 8 USDA macronutrient probes from external nutrient data. Chem (green) leads on every probe except `usda_energy_kcal`, with the consistent ordering Cooc < Core < Chem across rows.

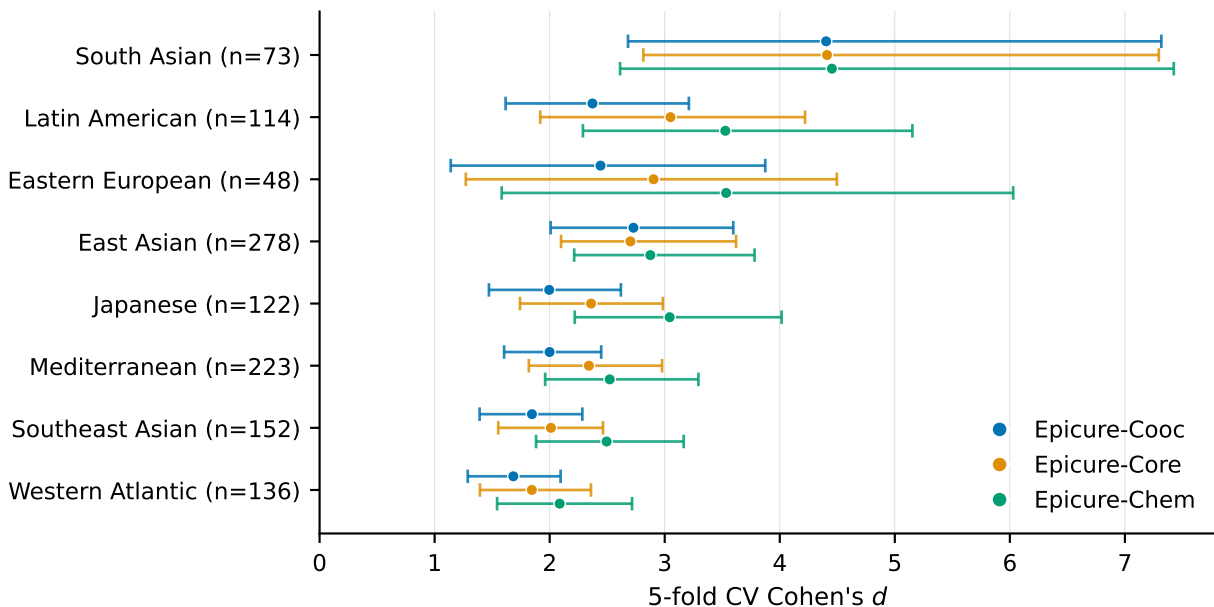


Figure 3: Per-region Cohen’s  $d$  (5-fold repeated CV, one-vs-rest on the distinctive-marker tags for each macro-region) for the three Epicure models, with 95% CIs.  $n$  is the number of tagged ingredients per region; higher  $d$  means more linearly separable. Regions are sorted by mean  $d$  across models. Chem leads on 8 of 8 regions; CIs widen sharply for low- $n$  regions (Eastern European, South Asian) but the cross-region ranking is consistent.

### 3.3 Emergent factors and modes

In order to discover the embedding’s natural axes without using any labels, we ran the multi-seed-stable FastICA + GMM mode-discovery pipeline of Section 2.5 on each Epicure model; the supplement’s *Factor Decomposition* appendix documents the factor-extraction method comparison, per-factor split-half stability, and a cuisine-orthogonalisation robustness check.

We found 20 stable factors per model and 150–200 modes per model (Cooc 150 modes across 41 properties; Core 193 / 44; Chem 200 / 43), each reading as a named culinary neighbourhood: *Sweet baking and dessert ingredients*, *South Asian whole spice blends*, *Mexican & Latin American Pantry*. Per-model mode listings are in the supplement’s *Mode Atlas* appendix. Figure 4 renders one representative factor per model as a worked example: the top-quartile of each factor is coloured by GMM-mode assignment on the model’s 2-D UMAP, with a short Claude-generated label at every mode’s centroid. We discuss factor indices as coordinates that locate modes rather than as named axes themselves; the interpretable culinary content lives at the mode level, not the factor pole.

We also found that emergent modes sit 5–6 $\times$  above the random-pair coherence baseline in every model: Figure 5 reports mean-cosine-to-pole of 0.611/0.833/0.703 for Cooc/Core/Chem against random-pair baselines of 0.097/0.348/0.115. The tightness margin (coherence – baseline) is comparable across the three models,  $\approx 0.5$ ; absolute coherence tracks each model’s overall concentration – Core’s PR = 94 pulls both pole tightness and the all-pairs floor upward, while the isotropic Cooc and Chem (PR  $\approx 174$  and 183) produce lower absolute coherence with the same margin. This means the unsupervised axes are not artefacts of a single seed and the modes that fall out of them are tight named neighbourhoods rather than arbitrary partitions – a vocabulary of navigation atoms alongside the supervised directions of Section 3.2. Section 4 demonstrates the operators that act on



Figure 4: One ICA factor per Epicure model with its GMM-mode decomposition, projected onto each model’s own 2-D UMAP. Coloured points are the top-quartile members of the highlighted factor, partitioned into GMM modes with Claude-generated labels at each mode’s median centroid; grey points are the rest of the 1,790-ingredient vocabulary. Each panel title carries a short Claude summary of the factor’s high-quartile derived from its  $K$  mode labels: a single named culinary identity when the modes cohere, or a description of the multi-cluster decomposition itself when they do not (here, all three picks fall in the decomposition regime – Cooc  $F_8$  splits into five distinct cuisine families, Core  $F_9$  into six savoury seasoning sub-clusters, Chem  $F_{14}$  into an East-Asian vs. Mediterranean savoury–sweet split, demonstrating that even compound-mediated metapaths surface multi-modal culinary geometry). ICA orientations are model-specific so the three factor indices do not correspond across panels. Full per-model factor summaries and per-mode atlases are in the supplement.

these atoms.

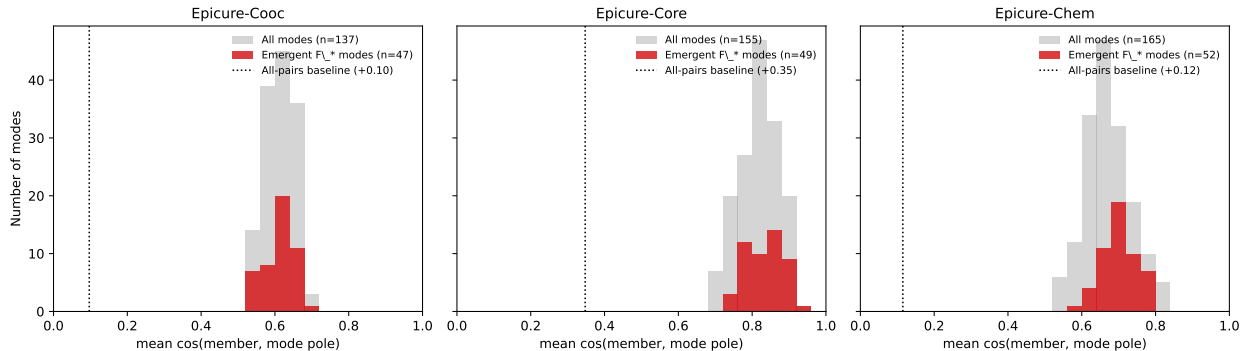


Figure 5: Distribution of per-mode coherence (mean cosine of members to mode pole) for each Epicure model. Red bars are emergent ( $F_*$ ) modes; gray bars are all modes including supervised properties for context; dotted line is the all-pairs random-pair baseline. Emergent modes sit well above baseline in every model: Cooc 0.611 vs. baseline 0.097, Core 0.833 vs. 0.348, Chem 0.703 vs. 0.115. The tightness margin (mode-coherence – baseline) is comparable across models ( $\approx 0.5$ ); absolute coherence tracks each model’s overall concentration (Core’s concentrated geometry pulls both pole tightness and the all-pairs baseline up; the isotropic Cooc and Chem sit lower in absolute terms with the same margin).

## 4 Transformations

The geometry of Section 3 – linearly recoverable supervised directions plus 150–200 named emergent modes per model – exposes two complementary operator families: nearest-neighbour *pairings* (Section 4.1) and SLERP-style *direction arithmetic* (Section 4.2) toward either a supervised direction or an emergent mode pole.

### 4.1 Pairings

In order to test how each embedding answers the simplest culinary question – what pairs with X – we compute the top-5 cosine-nearest neighbours plus the closest emergent mode (cosine to mode pole) for twelve canonical probe seeds, with FlavorGraph as an external foil.

We found that the three Epicure models return culinarily coherent peers at consistent granularity while FlavorGraph returns long preparation-level strings that fragment the co-occurrence signal across scattered food (and non-food) vocabulary (Table 6).



Table 6: Top-5 nearest neighbours by cosine similarity. The FlavorGraph column is retained as an illustrative foil – its 6,653-ingredient vocabulary contains brand names and preparation modifiers (e.g. “sourdough roll”, “kraft shredded triple cheddar cheese with a touch of philadelphia”) that our 1,790-canonical vocabulary collapses, so FG’s neighbours are drawn from a structurally different name space.

| Ingredient | Epicure-Cooc top-5                                                                                | Epicure-Core top-5                                                                            | Epicure-Chem top-5                                                                               | FlavorGraph top-5                                                                                                                                                                             |
|------------|---------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| chicken    | garlic (0.39), onion (0.37), black pepper (0.36), turkey (0.35), carrot (0.34)                    | pork (0.58), beef (0.57), chicken broth (0.55), peanut (0.52), cream of chicken soup (0.52)   | beef (0.41), pork (0.34), cream of chicken soup (0.31), buffalo wing sauce (0.29), peanut (0.28) | sourdough roll (0.83), macaroni shells and cheese (0.83), kraft shredded triple cheddar cheese with a touch of philadelphia (0.82), heads of garlic (0.81), montreal chicken seasoning (0.79) |
| salmon     | wasabi (0.35), avocado (0.32), ponzu (0.32), balsamic vinegar (0.32), scallop (0.32)              | trout (0.65), ham (0.63), mirin (0.62), cod (0.62), fish roe (0.61)                           | ham (0.50), trout (0.36), tuna (0.34), cod (0.33), miso (0.31)                                   | lemon pepper seasoning (0.91), wasabi paste (0.85), dried dill weed (0.84), liquid smoke flavoring (0.74), dried dill (0.73)                                                                  |
| tomato     | onion (0.48), parsley (0.47), garlic (0.46), olive oil (0.46), bell pepper (0.44)                 | bell pepper (0.61), red pepper (0.60), olive oil (0.59), onion (0.59), red onion (0.58)       | bean (0.29), bell pepper (0.29), red pepper (0.29), tortilla (0.28), onion (0.28)                | crisp salad green (0.90), dry chili pepper (0.89), light balsamic vinaigrette salad dressing (0.89), western salad dressing (0.89), tex mex cheese (0.89)                                     |
| basil      | parsley (0.45), olive oil (0.44), parmesan cheese (0.44), black pepper (0.42), white wine (0.39)  | oregano (0.71), tarragon (0.68), rosemary (0.67), olive oil (0.67), pasta (0.67)              | tarragon (0.46), oregano (0.46), rosemary (0.44), pasta (0.40), fennel (0.38)                    | light mozzarella cheese (0.78), minced garlic clove (0.75), italian plum tomato (0.74), italian cut green bean (0.73), fat free parmesan cheese (0.72)                                        |
| chocolate  | cocoa powder (0.55), vanilla (0.49), milk chocolate (0.48), almond (0.48), white chocolate (0.47) | cocoa powder (0.85), toffee (0.79), frosting (0.77), fudge (0.74), cream (0.73)               | toffee (0.69), cocoa powder (0.68), frosting (0.67), cream (0.65), marshmallow (0.64)            | pen (0.84), mint extract (0.82), espresso powder (0.81), tea biscuit (0.80), dark cocoa (0.71)                                                                                                |
| rice       | carrot (0.34), okra (0.33), vegetable oil (0.33), pea (0.31), dashi (0.31)                        | shiitake mushroom (0.60), nori (0.59), bok choy (0.58), brown rice (0.58), sesame seed (0.57) | caviar (0.31), nori (0.30), wheat (0.30), millet (0.30), puzi leaf (0.29)                        | frozen peas and carrot (0.93), boneless skinless chicken (0.82), chicken giblet (0.77), dark soya sauce (0.76), chicken part (0.71)                                                           |

*continued on next page*

Table 6 continued

| <b>Ingredient</b> | <b>Epicure-Cooc top-5</b>                                                                                      | <b>Epicure-Core top-5</b>                                                                                | <b>Epicure-Chem top-5</b>                                                                 | <b>FlavorGraph top-5</b>                                                                                                              |
|-------------------|----------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| butter            | cream (0.46), egg yolk (0.43), flour (0.41), milk (0.41), almond (0.38)                                        | milk (0.60), egg yolk (0.57), egg (0.56), parmesan cheese (0.50), white chocolate (0.49)                 | milk (0.48), flour (0.35), egg (0.33), chocolate (0.33), parmesan cheese (0.31)           | almond macaroon (0.86), snow crab leg (0.85), ginger in syrup (0.85), caramel square (0.85), yellow cling peach (0.84)                |
| soy sauce         | sesame oil (0.54), shiitake mushroom (0.47), ginger (0.47), oyster sauce (0.46), light soy sauce (0.45)        | sesame oil (0.75), light soy sauce (0.71), oyster sauce (0.70), shaoxing wine (0.70), doubanjiang (0.69) | scallion (0.45), enoki mushroom (0.35), bamboo shoot (0.32), sake (0.32), bok choy (0.32) | red rice (0.92), ginkgo nut (0.72), rice cake (0.58), chinese mustard (0.54), guava (0.52)                                            |
| lemon             | orange (0.42), mint (0.41), rosemary (0.40), cherry (0.40), mustard (0.38)                                     | lime (0.65), orange (0.62), thyme (0.58), fennel (0.57), olive oil (0.57)                                | lime (0.49), clementine (0.44), orange (0.43), pomegranate (0.38), parsley (0.37)         | white fleshed fish (0.88), lemon sherbet (0.88), rockfish fillet (0.88), capers in brine (0.87), baby chicken (0.87)                  |
| cumin             | garlic (0.44), coriander (0.43), bay leaf (0.43), chili powder (0.42), black pepper (0.39)                     | chili powder (0.74), coriander (0.70), turmeric (0.70), chickpea (0.67), green chili (0.67)              | turmeric (0.45), ajwain (0.41), black pepper (0.40), parsley (0.40), chili powder (0.40)  | hatch chile (0.87), mexican tomato sauce (0.81), dark mexican beer (0.79), mexican style diced tomato (0.79), green taco sauce (0.78) |
| shrimp            | chili pepper (0.36), sesame oil (0.36), clam (0.36), cabbage (0.35), squid (0.34)                              | squid (0.67), clam (0.67), oyster (0.62), crab stick (0.62), crab (0.60)                                 | clam (0.43), oyster (0.38), squid (0.37), krill (0.36), crab (0.31)                       | firm white fish (0.76), cooked grit (0.76), quick cooking grit (0.73), grit (0.73), old bay seasoning (0.64)                          |
| lentil            | turmeric (0.35), cayenne pepper (0.32), mustard seed (0.31), pomegranate molasses (0.31), sambar powder (0.31) | chickpea (0.79), vegetable stock (0.73), butternut squash (0.69), olive oil (0.68), zucchini (0.68)      | chickpea (0.61), butternut squash (0.54), quinoa (0.49), zucchini (0.49), kale (0.49)     | fresh bay leaf (0.75), dried garbanzo bean (0.75), whole bay leaf (0.72), dried navy bean (0.64), long grain brown rice (0.62)        |

We also found that the three Epicure models retrieve different *kinds* of neighbour for the same seed. For chicken, Cooc’s top hit is *garlic* (recipe companion) and its full top-5 is mostly aromatic vegetables (*garlic, onion, black\_pepper, turkey, carrot*); Core’s top hit is *pork* (chemistry peer); Chem’s is *beef*, and its top-5 sits in the chicken-cooking neighbourhood – two protein peers plus three canonical chicken accompaniments (*beef, pork, cream\_of\_chicken\_soup, buffalo\_wing\_sauce, peanut*). For basil, Cooc retrieves *parsley* (co-occurrence peer); Core *oregano* and Chem *tarragon* both sit in the Italian-herb chemistry cluster and share four of five top-5 peers (*oregano, tarragon, rosemary, pasta*), while Cooc’s top-5 reaches for basil’s pasta-pantry context (*olive\_oil, parmesan\_cheese, black\_pepper, white\_wine*). This means the three siblings expose the two paths a chef might take when reaching for a replacement: “what else do I cook with this” (Cooc) versus “what shares its flavour profile” (Core and Chem).

**Mode-membership pairings.** Table 7 extends the simple top- $K$  neighbour view to mode-membership lookup: for a probe seed, the closest emergent mode in each model (cosine to mode pole) together with the other top members of that mode. This separates “where in the atlas does the seed live” from “what’s nearest to the seed” – a chef-facing tool typically wants both.

Table 7: Emergent pairings catalogue: for each probe seed, the closest emergent ( $F_*$ ) mode the seed sits in for each model, with the other top members of that mode. The sweet-confection neighbourhood for chocolate is shared across all three models, while cuisine framing for savoury and umami seeds (tomato, miso, preserved lemon) shifts across the chemistry/co-occurrence axis.

| Seed            | Cooc                                                                                                                       | Core                                                                                                                             | Chem                                                                                                                 |
|-----------------|----------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| chocolate       | F_4/M3<br><b>chocolate and coffee confections</b> (cos=+0.69)<br>ganache<br>coffee liqueur<br>cocoa powder                 | F_15/M0<br><b>American confectionery and sweet treats</b> (cos=+0.86)<br>graham cracker<br>toffee<br>fudge                       | F_5/M5<br><b>confectionery and dessert components</b> (cos=+0.84)<br>gelatin<br>cream<br>meringue                    |
| tomato          | F_8/M0<br><b>Everyday Western savory vegetables</b> (cos=+0.68)<br>black pepper<br>onion<br>lettuce                        | F_0/M0<br><b>Mediterranean savory cooking staples</b> (cos=+0.67)<br>vegetable stock<br>garlic<br>zucchini                       | F_5/M1<br><b>Hearty bean and sausage stew ingredients</b> (cos=+0.32)<br>red onion<br>cannellini bean<br>bell pepper |
| miso            | F_10/M1<br><b>Japanese hot pot ingredients</b> (cos=+0.44)<br>enoki mushroom<br>garland chrysanthemum<br>shiitake mushroom | F_8/M5<br><b>Japanese vegetables and umami seasonings</b> (cos=+0.81)<br>rice vinegar<br>enoki mushroom<br>oyster mushroom       | F_13/M4<br><b>Savory protein-rich seafood and cheese</b> (cos=+0.55)<br>cheese<br>cod<br>octopus                     |
| lentil          | F_2/M2<br><b>Mediterranean pantry seeds and aromatics</b> (cos=+0.40)<br>pomegranate<br>salt<br>pumpkin seed               | F_12/M3<br><b>Mediterranean whole-food vegetables and grains</b> (cos=+0.79)<br>balsamic vinegar<br>chicken broth<br>fennel seed | F_7/M4<br><b>Mediterranean garden vegetables and grains</b> (cos=+0.58)<br>kefir<br>beet<br>semolina                 |
| preserved lemon | F_2/M2<br><b>Mediterranean pantry seeds and aromatics</b> (cos=+0.29)<br>pomegranate<br>salt<br>pumpkin seed               | F_14/M0<br><b>Middle Eastern spice blends and staples</b> (cos=+0.58)<br>berbere<br>falafel<br>baharat                           | F_14/M4<br><b>Mediterranean savory pantry staples</b> (cos=+0.29)<br>olive oil<br>cayenne pepper<br>balsamic vinegar |

*continued on next page*

Table 7 continued

| Seed   | Cooc                                                                                                 | Core                                                                                                                      | Chem                                                                                                            |
|--------|------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| coffee | F_6/M2<br>sweet cocktail and confection ingredients (cos=+0.68)<br>rum<br>white chocolate<br>ganache | F_19/M3<br>Sweet dessert liqueurs and confections (cos=+0.58)<br>coffee liqueur<br>chocolate liqueur<br>maraschino cherry | F_13/M3<br>Sweet liqueurs and cocktail ingredients (cos=+0.45)<br>coffee liqueur<br>liqueur<br>hazelnut liqueur |

Nearest neighbours expose where the seed already sits; steering the seed in a chosen direction requires an explicit operator, which the next subsection introduces.

## 4.2 Direction arithmetic

In order to test whether a seed can be steered along a culinary axis – and how cleanly that motion respects either supervised labels or unsupervised mode geometry – we apply SLERP rotation of the seed toward a unit direction by angle  $\theta$  on the unit sphere. At  $0^\circ$  the rotated query is the unmodified seed; at  $60^\circ$  its cosine similarity to the seed has dropped to 0.5 and the target’s neighbourhood dominates. Two direction families are available: *supervised* pole vectors built from labelled tags (cuisine macro-regions, food groups, NOVA processing class), and the *emergent* factor-mode poles from Section 3.3.

**SLERP toward supervised directions.** Table 8 reports four hero rotations toward supervised pole vectors at  $30^\circ$  and  $60^\circ$ . We found that the destinations are label-aligned in every model: rice rotated toward the South-Asian direction at  $30^\circ$  retrieves *curry leaf*, *masoor dal*, *urad dal*, *chana dal*, *fenugreek seed* in Cooc; corn rotated toward Latin American at  $30^\circ$  retrieves *salsa verde*, *tomatillo*, *queso fresco*, *fajita seasoning*, *corn tortilla*. Multi-constraint queries – chicken rotated toward processed + Western Atlantic at  $60^\circ$  – converge on mid-century American home-cooking staples (*swiss cheese*, *steak sauce*, *turkey*, *sour cream*, *ranch dressing* in Cooc; *cheddar cheese*, *cream of chicken soup*, *crescent roll*, *alfredo sauce*, *ranch dressing* in Core; *colby cheese*, *buffalo wing sauce*, *ranch dressing*, *cream of chicken soup*, *alfredo sauce* in Chem). This means supervised SLERP is a predictable, label-aligned steering operator across all three siblings.

Table 8: Direction arithmetic (SLERP), hero cases for the main paper. Seed is rotated toward the learned direction on the unit sphere by the specified angle; top-5 nearest neighbours of the rotated query vector are reported.  $0^\circ$  is the unmodified seed,  $60^\circ$  is a full rotation (cosine similarity to seed = 0.50). See `direction_arithmetic_full` for all 48 test cases.

| Test case             | Rotation   | Epicure-Cooc top-5 | Epicure-Core top-5 | Epicure-Chem top-5 |
|-----------------------|------------|--------------------|--------------------|--------------------|
| rice + South_Asian    | $0^\circ$  | carrot             | shiitake mushroom  | caviar             |
|                       |            | okra               | nori               | nori               |
|                       |            | vegetable oil      | bok choy           | wheat              |
|                       |            | pea                | sesame seed        | millet             |
|                       |            | dashi              | pea                | puzi leaf          |
|                       | $30^\circ$ | curry leaf         | chana dal          | chana dal          |
|                       |            | masoor dal         | fenugreek leaf     | toor dal           |
|                       |            | urad dal           | urad dal           | fenugreek seed     |
|                       |            | chana dal          | toor dal           | kashmiri chili     |
|                       |            | fenugreek seed     | horse gram         | sambar powder      |
|                       | $60^\circ$ | curry leaf         | chana dal          | chana dal          |
|                       |            | chana dal          | toor dal           | kashmiri chili     |
|                       |            | urad dal           | kashmiri chili     | toor dal           |
|                       |            | masoor dal         | urad dal           | fenugreek seed     |
|                       |            | horse gram         | horse gram         | amchur             |
| corn + Latin_American | $0^\circ$  | pea                | bell pepper        | tortilla           |
|                       |            | scallion           | potato             | guascas            |
|                       |            | shrimp             | carrot             | red pepper         |
|                       |            | red onion          | pinto bean         | oregano            |
|                       |            | rice               | cheddar cheese     | epazote            |
|                       | $30^\circ$ | salsa verde        | tomatillo          | corn tortilla      |
|                       |            | tomatillo          | corn tortilla      | queso fresco       |
|                       |            | queso fresco       | epazote            | epazote            |
|                       |            | fajita seasoning   | salsa              | salsa              |
|                       |            | corn tortilla      | nopal              | enchilada sauce    |
|                       | $60^\circ$ | tomatillo          | tomatillo          | corn tortilla      |
|                       |            | queso fresco       | corn tortilla      | queso fresco       |
|                       |            | corn tortilla      | nopal              | salsa              |
|                       |            | salsa verde        | queso fresco       | guajillo chile     |
|                       |            | refried beans      | epazote            | poblano pepper     |

*continued on next page*



Table 8 continued

| Test case                                 | Rotation | Epicure-Cooc top-5                                                    | Epicure-Core top-5                                                                          | Epicure-Chem top-5                                                                             |
|-------------------------------------------|----------|-----------------------------------------------------------------------|---------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| chicken + processed +<br>Western_Atlantic | 0°       | garlic<br>onion<br>black pepper<br>turkey<br>carrot                   | pork<br>beef<br>chicken broth<br>peanut<br>cream of chicken soup                            | beef<br>pork<br>cream of chicken soup<br>buffalo wing sauce<br>peanut                          |
|                                           | 30°      | turkey<br>swiss cheese<br>onion<br>steak sauce<br>sour cream          | cheddar cheese<br>cream of chicken soup<br>alfredo sauce<br>beef<br>turkey                  | cream of chicken soup<br>buffalo wing sauce<br>beef<br>ranch dressing<br>cheddar cheese        |
|                                           | 60°      | swiss cheese<br>steak sauce<br>turkey<br>sour cream<br>ranch dressing | cheddar cheese<br>cream of chicken soup<br>crescent roll<br>alfredo sauce<br>ranch dressing | colby cheese<br>buffalo wing sauce<br>ranch dressing<br>cream of chicken soup<br>alfredo sauce |
| bread + high protein                      | 0°       | olive oil<br>parsley<br>paprika<br>thyme<br>anchovy                   | pasta<br>artichoke<br>olive oil<br>parsley<br>red onion                                     | pasta<br>caper<br>artichoke<br>anchovy<br>provolone cheese                                     |
|                                           | 30°      | olive oil<br>parsley<br>black pepper<br>pasta<br>paprika              | olive oil<br>pasta<br>artichoke<br>red onion<br>anchovy                                     | pasta<br>artichoke<br>provolone cheese<br>anchovy<br>caper                                     |
|                                           | 60°      | black pepper<br>pasta<br>tomato<br>parmesan cheese<br>pepperoni       | anchovy<br>pita bread<br>olive oil<br>tuna<br>sausage                                       | provolone cheese<br>monkfish<br>anchovy<br>hot dog<br>pita bread                               |

**SLERP toward emergent mode poles.** The same SLERP operator works on emergent targets. Table 9 reports three rotations from various seeds toward an *intent* – a target concept resolved per model to its best-matching mode by label keyword. The target  $(F_X, M_Y)$  coordinate differs across models because ICA orientations are model-specific; the cells show the actual coordinate used and the top-5 hits. We found that the destinations differ across models in ways that mirror their geometry. *chocolate* rotated toward sweet baking lands on a baking-and-confection cluster in all three models, though the cultural framing differs: Cooc and Core both reach a Western sweet-baking neighbourhood (*cocoa powder, vanilla, coffee* for Cooc; *baking powder, chia seed, whole wheat flour* for Core), while Chem lands on an East-Asian dessert mode anchored by *red bean paste, matcha powder, purple sweet potato*. *chicken* rotated toward Southeast-Asian aromatics traces the same chemistry/co-occurrence split: Cooc picks an Indonesian spice-paste mode (*candlenut, kencur, garam masala*), Core a broader East/Southeast-Asian pantry mode (*rice noodle, bean sprout, fish ball*), and Chem a Southeast-Asian chili-spice mode (*chili pepper, sichuan peppercorn, birds eye chili*). *tomato* rotated toward a Mediterranean savoury pantry retrieves model-specific regional cuts of the same concept: a savoury whole-food Mediterranean staples mode in Cooc (*turkey, butternut squash, kale*), an Eastern Mediterranean cheese-and-flatbread mode in Core (*tulum cheese, kasseri cheese, yufka*), and a Caucasian–Mediterranean pantry mode in Chem (*sulguni cheese, sun dried tomato, adjika*). This means emergent SLERP exposes each model’s training bias – Cooc reaching recipe-context neighbours, Chem reaching chemistry-clustered ones – as a navigable knob rather than hiding it.

Table 9: Hero *mode\_rotate* examples by SHARED INTENT, not shared coordinate. Each row pins a target concept (e.g. *sweet baking / confection*) and resolves it to the model-specific mode whose label best matches; the table row then rotates the seed toward THAT model’s mode, showing the actual (factor, mode\_id) coordinate used and the resulting top-5 hits. Rows therefore compare same-concept rotations across the three Epicure models, not same-coordinate rotations. The angle is 60° throughout. See Table 10 for an angle sweep showing how the seed identity fades as the angle grows.

| Seed      | Intent                    | Cooc                                                                                                                                                              | Core                                                                                                                                                                                         | Chem                                                                                                                                                                                                  |
|-----------|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| chocolate | Sweet baking / confection | F_7/M1<br><b>Sweet baking and plant-based pantry staples</b><br>cocoa powder (+0.66)<br>vanilla (+0.65)<br>coffee (+0.65)<br>hazelnut (+0.64)<br>cacao (+0.63)    | F_5/M2<br><b>Sweet baking and confectionery ingredients</b><br>baking powder (+0.81)<br>chia seed (+0.81)<br>whole wheat flour (+0.78)<br>baking soda (+0.77)<br>soy protein isolate (+0.76) | F_12/M1<br><b>East Asian confectionery and sweet baking ingredients</b><br>red bean paste (+0.72)<br>matcha powder (+0.71)<br>purple sweet potato (+0.69)<br>mochi (+0.66)<br>mung bean paste (+0.63) |
| chicken   | Southeast-Asian aromatics | F_0/M0<br><b>Indonesian spice paste aromatics</b><br>candlenut (+0.66)<br>kencur (+0.65)<br>garam masala (+0.63)<br>palm sugar (+0.61)<br>sweet soy sauce (+0.60) | F_13/M0<br><b>East and Southeast Asian pantry staples</b><br>rice noodle (+0.89)<br>bean sprout (+0.88)<br>fish ball (+0.88)<br>udon noodle (+0.87)<br>sesame oil (+0.87)                    | cf_minty/M4<br><b>Southeast Asian aromatic chili spices</b><br>chili pepper (+0.73)<br>sichuan peppercorn (+0.72)<br>birds eye chili (+0.67)<br>shaoxing wine (+0.64)<br>dark soy sauce (+0.63)       |

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Table 9 continued

| Seed   | Intent                         | Cooc                                                                                                                                                                                               | Core                                                                                                                                                                             | Chem                                                                                                                                                                                            |
|--------|--------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| tomato | Mediterranean<br>savory pantry | F_7/M2<br><b>Savory whole-food<br/>Mediterranean pantry<br/>staples</b><br>turkey (+0.63)<br>butternut squash (+0.62)<br>kale (+0.62)<br>vegetable stock (+0.61)<br>portobello mushroom<br>(+0.61) | F_1/M1<br><b>Eastern Mediterranean<br/>pantry staples</b><br>tulum cheese (+0.87)<br>kasseri cheese (+0.85)<br>yufka (+0.83)<br>ezine cheese (+0.82)<br>kashkaval cheese (+0.81) | F_7/M1<br><b>Caucasian and Eastern<br/>Mediterranean pantry<br/>staples</b><br>sulguni cheese (+0.66)<br>sun dried tomato (+0.65)<br>adjika (+0.63)<br>khmeli suneli (+0.63)<br>bryndza (+0.63) |

**The angle is a continuous knob.** Table 10 demonstrates how the rotated query transitions from seed-dominated to target-dominated as the angle grows. Two seeds (chicken and beef) rotate toward a single canonical chef intent – the *Mexican / Tex-Mex pantry* mode (chicken fajitas / beef barbacoa territory) – at three angles (0°, 30°, 60°) in each Epicure model. We found that at 0° the rotated query is the unmodified seed and the top-5 is the seed’s own nearest neighbourhood (Cooc beef returns *onion, pork, black\_pepper, garlic, potato*; Core chicken returns *pork, beef, chicken\_broth, peanut, cream\_of\_chicken\_soup*); by 30° Tex-Mex intermediates dominate (Cooc beef: *corn\_tortilla, monterey\_jack\_cheese, onion, pinto\_bean, salsa*; Core chicken: *monterey\_jack\_cheese, flour\_tortilla, corn\_tortilla, salsa\_verde, enchilada\_sauce*); at 60° both seeds collapse onto a nearly identical Mexican-specialty neighbourhood – in Core both retrieve the same Tex-Mex top-5 (*corn\_tortilla, salsa, monterey\_jack\_cheese, flour\_tortilla, tortilla*); in Cooc both share *corn\_tortilla, monterey\_jack\_cheese, salsa\_verde, salsa, poblano\_pepper*; in Chem both share *poblano\_pepper, salsa, cotija\_cheese, corn\_tortilla, monterey\_jack\_cheese*. The 60° destinations are specialty Mexican ingredients (*cotija\_cheese, ancho\_chile, poblano\_pepper, salsa\_verde*) the seeds themselves do not retrieve directly; the rotation surfaces them from a generic meat seed. This means the angle is a continuous dial between seed and target, and chef-facing tools should expose it so a user can stay close to the seed when refining or travel further when exploring.

Table 10: Angle sweep on *mode\_rotate*: chicken and beef rotated toward the *Mexican / Tex-Mex pantry* intent in each Epicure model at three angles (0°, 30°, 60°). At 0° the rotated query is the unmodified seed and the top-5 is the seed’s own nearest neighbourhood; as the angle grows the query moves toward the target pole and at 60° the two seeds collapse onto a nearly identical target neighbourhood, demonstrating that the angle is a continuous knob between *stay near the seed* and *go to the target*. All other rows in Table 9 use 60°.

| Seed                 | Angle | Cooc                                                                                        | Core                                                                                                        | Chem                                                                                                             |
|----------------------|-------|---------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|
| <i>Seed: chicken</i> |       |                                                                                             |                                                                                                             |                                                                                                                  |
| chicken              | 0°    | garlic (+0.39)<br>onion (+0.37)<br>black pepper (+0.36)<br>turkey (+0.35)<br>carrot (+0.34) | pork (+0.58)<br>beef (+0.57)<br>chicken broth (+0.55)<br>peanut (+0.52)<br>cream of chicken soup<br>(+0.52) | beef (+0.41)<br>pork (+0.34)<br>cream of chicken soup<br>(+0.31)<br>buffalo wing sauce (+0.29)<br>peanut (+0.28) |

continued on next page

Table 10 continued

| Seed              | Angle | Cooc                                                                                                                      | Core                                                                                                                              | Chem                                                                                                                      |
|-------------------|-------|---------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| chicken           | 30°   | corn tortilla (+0.54)<br>salsa (+0.51)<br>monterey jack cheese (+0.50)<br>fajita seasoning (+0.50)<br>salsa verde (+0.50) | monterey jack cheese (+0.80)<br>flour tortilla (+0.79)<br>corn tortilla (+0.79)<br>salsa verde (+0.77)<br>enchilada sauce (+0.77) | enchilada sauce (+0.51)<br>salsa (+0.51)<br>flour tortilla (+0.51)<br>corn tortilla (+0.51)<br>salsa verde (+0.49)        |
| chicken           | 60°   | corn tortilla (+0.67)<br>monterey jack cheese (+0.66)<br>salsa verde (+0.65)<br>salsa (+0.63)<br>poblano pepper (+0.63)   | corn tortilla (+0.91)<br>salsa (+0.91)<br>monterey jack cheese (+0.89)<br>flour tortilla (+0.89)<br>tortilla (+0.89)              | corn tortilla (+0.69)<br>poblano pepper (+0.68)<br>salsa (+0.68)<br>cotija cheese (+0.67)<br>ancho chile (+0.67)          |
| <i>Seed: beef</i> |       |                                                                                                                           |                                                                                                                                   |                                                                                                                           |
| beef              | 0°    | onion (+0.41)<br>pork (+0.37)<br>black pepper (+0.35)<br>garlic (+0.35)<br>potato (+0.32)                                 | chicken (+0.57)<br>taco seasoning (+0.56)<br>bell pepper (+0.55)<br>enchilada sauce (+0.55)<br>mushroom (+0.54)                   | chicken (+0.41)<br>taco sauce (+0.34)<br>taco seasoning (+0.33)<br>enchilada sauce (+0.30)<br>mushroom (+0.30)            |
| beef              | 30°   | corn tortilla (+0.52)<br>monterey jack cheese (+0.52)<br>onion (+0.48)<br>pinto bean (+0.48)<br>salsa (+0.47)             | salsa (+0.81)<br>flour tortilla (+0.81)<br>monterey jack cheese (+0.81)<br>enchilada sauce (+0.80)<br>corn tortilla (+0.79)       | enchilada sauce (+0.55)<br>salsa (+0.53)<br>pinto bean (+0.52)<br>monterey jack cheese (+0.51)<br>flour tortilla (+0.49)  |
| beef              | 60°   | corn tortilla (+0.67)<br>monterey jack cheese (+0.67)<br>salsa verde (+0.63)<br>poblano pepper (+0.62)<br>salsa (+0.62)   | corn tortilla (+0.91)<br>salsa (+0.90)<br>monterey jack cheese (+0.89)<br>flour tortilla (+0.89)<br>tortilla (+0.89)              | poblano pepper (+0.70)<br>salsa (+0.68)<br>cotija cheese (+0.68)<br>corn tortilla (+0.67)<br>monterey jack cheese (+0.67) |

Supervised SLERP gives label-aligned steering; emergent SLERP gives steering without curated labels; the angle is a continuous dial between seed and target. Section 5 considers what corpus and operator extensions these primitives suggest.

## 5 Discussion

### 5.1 What the controlled comparison shows

Cooc, Core, and Chem share architecture, hyperparameters, vocabulary, graph node set, and the entire 203,508-edge co-occurrence backbone (Section 2); they differ only in which typed walks the skip-gram objective sees and at what rate. Two findings follow from holding everything else fixed. First, the  $\text{Cooc} < \text{Core} < \text{Chem}$  ordering of supervised direction quality (Section 3.2) holds on every probe stratum we test, including the five basic-taste, eight USDA-macronutrient, and eight cuisine-macro-region probes that the compound-feature schema never sees. Chemistry-mediated walks therefore act as a structural prior whose reach extends beyond the labels they directly encode: routing context through shared aroma compounds makes a broader family of culinary concepts linearly recoverable than the schema names, and Mikolov-style linear directions [Mikolov et al., 2013] are the mechanism by which that prior becomes geometry. Second, Core’s concentrated geometry (participation ratio 94.2 against Cooc’s 173.6 and Chem’s 183.1; Section 3.1) is a deliberate

consequence of the  $10\times$  I-I walk injection, not a corpus-induced collapse of the kind [Mu et al. \[2017\]](#) address. It coincides with stronger linear probes than either isotropic sibling and with the tightest emergent modes of the three (Section 3.3), so the concentration is a design lever rather than a defect to rescue.

## 5.2 From recommendation to navigation

The chemistry-vs-recipe-context axis surfaces twice in the operator output of Section 4: at the nearest-neighbour level the same seed returns a recipe companion under Cooc and a flavour-profile peer under Chem, and at the SLERP-destination level the same seed and target angle land on culturally different framings of the target concept depending on the sibling. The user-facing primitives therefore decompose into three independent choices, all expressed on the same 300-D embedding: which sibling to query (which question is being asked, co-occurrence companion or flavour-profile peer), which direction to rotate toward (a supervised pole vector or an emergent factor-mode pole), and how far to travel (the SLERP angle). Closest-mode lookup (Section 4.1) gives users the named-cluster query a knowledge graph like FoodKG [[Hausmann et al., 2019](#)] would offer (*which named region is this ingredient in?*) without sacrificing the continuous-geometry query that an embedding like FlavorGraph [[Park et al., 2021](#)] is designed for; the two affordances live on the same 300-D model rather than in separate systems. The methodological move that makes this possible – treating the walk schema as a named axis rather than an architectural constant – applies to any future fusion of chemistry, nutrient, sensory, image, or recipe-text signals.

## 5.3 Limitations

**Corpus imbalance.** The 4.14M-recipe corpus is roughly half East Asian and a tenth Mediterranean, with single-digit shares for South Asian, Eastern European, and Latin American cuisines (Section 2.1). The held-out-cuisine  $d$  confidence intervals (Figure 3) widen accordingly in the smaller regions; the cross-region *ranking* of the three siblings is nevertheless stable, so the imbalance limits resolution within a region more than it threatens the synthesis above.

**Hub coverage.** 525 of 1,790 canonical ingredients anchor against FlavorDB under our entity-unique matching policy (523 retain active I-C edges after the `min_compound_degree=2` filter); the remaining 1,267 non-hubs participate in both Core and Chem, but they reach compound context only indirectly, through the via-compound metapath `N-H-C[x]-H-N` (Section 2.4) that bridges two non-hubs through a hub-compound-hub spine and contributes the bulk of Chem’s skip-gram pair budget. Their chemistry signal is therefore one walk-hop further removed from the compound vertex than that of the 523 hubs; broader compound coverage (FoodDB, USDA Food Patterns Equivalents) would promote more non-hubs to hub status and shorten that chain.

**LLM dependence in the pipeline.** Canonicalisation, cuisine tagging, and the factor/mode label generation all use Claude under deterministic decoding, and every LLM-touched output is logged and inspectable. The embeddings themselves are LLM-free – the skip-gram objective sees only walk sequences over canonical ingredient and compound tokens – so the geometry we analyse is not directly conditioned on LLM judgements, but the canonical vocabulary that defines its node set is.

## 6 Conclusions

Computational gastronomy has moved from the descriptive flavour network of Ahn et al. [2011], through compound catalogues (FlavorDB [Garg et al., 2017], FooDB [The Metabolomics Innovation Centre, 2020]) and integrated knowledge graphs (FoodKG [Hausmann et al., 2019]), to distributed-representation food embeddings typified by FlavorGraph [Park et al., 2021]. Epicure suggests the next step is to expose the operators that act on such an embedding: a 300-D vector becomes useful to a chef when it is wrapped in nearest-neighbour pairings, closest-mode lookup, and SLERP rotation by a continuous angle, and when the inductive biases inside it are exposed as named, controllable axes rather than hidden in the choice of network. Three openings extend the work directly: a continuous mixing parameter at the walker that would turn the three siblings into a parameterised family and let the chemistry-vs-recipe-context trade-off be tuned rather than chosen; a richer set of operators beyond single mode jumps – intra-mode interpolation, multi-direction blends, and constrained traversal (*rotate toward Mediterranean but stay in the dairy mode*); and cross-modal grounding through the shared canonical vocabulary, so the SLERP operator can cross from ingredient space into recipe-text, image, or sensory-descriptor space on the same model. More broadly, the methodological move of treating the walk schema as the experimental variable applies to any future fusion of culinary signals. The next concrete artefact is a single chef-facing interface that exposes all three controls – model choice (Cooc/Core/Chem), closest-mode lookup, and the SLERP angle – in one place; measuring what real users do with that interface is the next empirical step.

## Declaration of Generative AI Use

This work used large language models in two capacities. **Data pipeline:** Anthropic Claude Opus family models (internal deployment IDs 4.6 and 4.7) [Anthropic, 2026a,b] performed all ingredient classification under deterministic decoding (temperature 0–0.1), including translation of non-English terms, canonical-vocabulary construction, dedup adjudication, 1:1 matching against USDA Food-Data Central and FlavorDB, cuisine-marker tagging, and generation of the sensory scores used as direction-quality ground truth. Google’s **gemini-embedding-001** endpoint [Google Cloud, 2026, Lee et al., 2025] was used to compute cosine similarity between canonical-name candidates during one dedup stage. All LLM outputs were validated by rule-based post-processing or human review. **Writing assistance:** Anthropic Claude Opus family models (internal deployment IDs 4.6 and 4.7) [Anthropic, 2026a] were used for drafting, editing, and code generation. All scientific claims, experimental design, and interpretations are the authors’ own.

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